Stock Price Forecasting Using Data From Yahoo Finance and Analyzing Seasonal and Nonseasonal Trend

Jeong Hoon Choi
USCID: 5023184813

DSCI-560
Data Science Professional Practicum

Time Series Analysis

- Variation in Trends (seasonal graph movement)

- Holt-Winters method(forecasting): three exponential smoothening to capture patterns

l: level, b: trend,

$$l_x = \alpha(y_x - s_x - L) + (1-\alpha)(l_x - 1 + b_x - 1) \quad (1)$$
$$b_x = \beta(l_x - l_{x-1}) + (1-\beta)b_{x-1} \quad (2)$$
$$s_x = \gamma(y_x - l_x) + (1-\gamma)s_x - L \quad (3)$$
$$y_{x+m} = l_{x+m} + b_x + s_x - L + 1 + (m-1) mod L \quad (4)$$

- Nonseasonal ARIMA: Auto Regression + Moving Average

ACF, PACF to decide parameters (p, d, q)

$$y'_t = c + \varphi_1 y'_{t-1} + \cdots + \varphi_p y'_{t-p} + \theta_1 e_{t-1} + \cdots + \theta_q e_{t-q} + e_t \quad (5)$$

time-seriese prediction
- variation in trends

(Holt-Winters
ARIMA : ACF, PACF → p, d, q) ⇒ APPLE from Yahoo Finance

Result & Observations

The close prices of Apple stock provided by Yahoo Finance from 2000 to 2018 were used for the analysis. (data from the first day of each month)

Jan, 2000 – Jan, 2016 is used as training data, and the data thereafter is used as test data.

Using two simpler models performs better than using a single model.

Question: Explain the ARIMA model.

Answer:

ARIMA (AutoRegressive Integrated Moving Average)

In ARIMA, p represents the number of past data points that influence the current time in the Auto Regression component. D represents the number of differencing steps required for stationarity. Q represents the number of past errors referenced in the Moving Average component.

(q is typically determined using the ACF and p is determined using the PACF.)